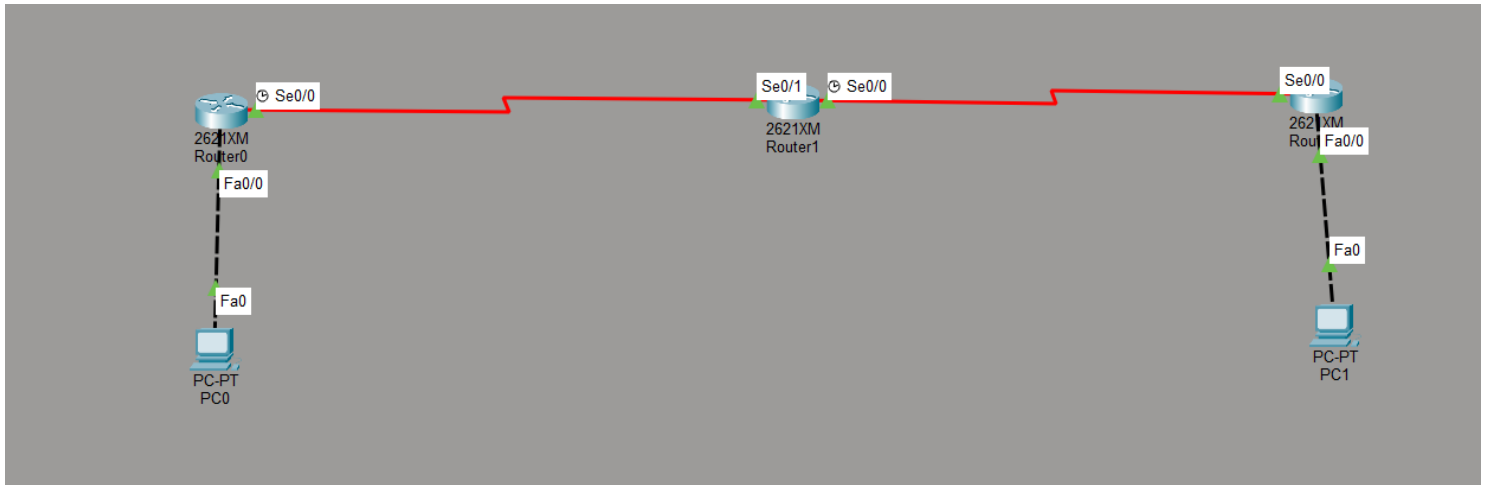


Q-Using Cisco Packet Tracer, create a network topology with three routers in two OSPF areas (Area 0 and Area 1), and assign appropriate IP addresses to all interfaces.

Topology



Router 0

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface Serial0/0
Router(config-if)#ip address 20.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up

Router(config-if)#
Router(config-if)#exit
Router(config)#router ospf 10
Router(config-router)#network 10.0.0.0 0.255.255.255 area 0
Router(config-router)#network 20.0.0.0 0.255.255.255 area 0
Router(config-router)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
00:06:30: %OSPF-5-ADJCHG: Process 10, Nbr 30.0.0.1 on Serial0/0 from LOADING to FULL, Loading Done

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
30.0.0.1	0	FULL/ -	00:00:38	20.0.0.2	Serial0/0

```
Router#
```

Router 1

```

Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface Serial0/0
Router(config-if)#ip address 30.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/1
Router(config-if)#ip address 20.0.0.2 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1, changed state to up

%LINK-5-CHANGED: Interface Serial0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up

Router(config-if)#exit
Router(config)#router ospf 10
Router(config-router)#
Router(config-router)#network 20.0.0.0 0.255.255.255 area 0
Router(config-router)#network 30.0.0.0 0.255.255.255 area 0
00:06:21: %OSPF-5-ADJCHG: Process 10, Nbr 20.0.0.1 on Serial0/1 from LOADING to FULL, Loading Done

Router(config-router)#network 30.0.0.0 0.255.255.255 area 0
Router(config-router)#no network 30.0.0.0 0.255.255.255 area 0
Router(config-router)#network 30.0.0.0 0.255.255.255 area 1
Router(config-router)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
00:07:34: %OSPF-5-ADJCHG: Process 10, Nbr 40.0.0.1 on Serial0/0 from LOADING to FULL, Loading Done

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip ospf neighbor

```

Neighbor ID	Pri	State	Dead Time	Address	Interface
20.0.0.1	0	FULL/ -	00:00:34	20.0.0.1	Serial0/1
40.0.0.1	0	FULL/ -	00:00:33	30.0.0.2	Serial0/0

```

Router#

```

Router 2

```

Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 40.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface Serial0/0
Router(config-if)#ip address
% Incomplete command.
Router(config-if)#ip address 30.0.0.2 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up

Router(config-if)#
Router(config-if)#exit
Router(config)#router ospf 10
Router(config-router)#network 30.0.0.0 0.255.255.255 area 1
Router(config-router)#
00:07:22: %OSPF-5-ADJCHG: Process 10, Nbr 30.0.0.1 on Serial0/0 from LOADING to FULL, Loading Done

Router(config-router)#network 40.0.0.0 0.255.255.255 area 1
Router(config-router)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip ospf neighbor

```

Neighbor ID	Pri	State	Dead Time	Address	Interface
30.0.0.1	0	FULL/ -	00:00:39	30.0.0.1	Serial0/0

```

Router#

```